

# **Homework4**

## **Glows.ai Environment Setup**

# Usage

- We utilize the computing resources provided by Glows AI.
- [Entrypoint](#)
- Upon first login, you are required to change your password.

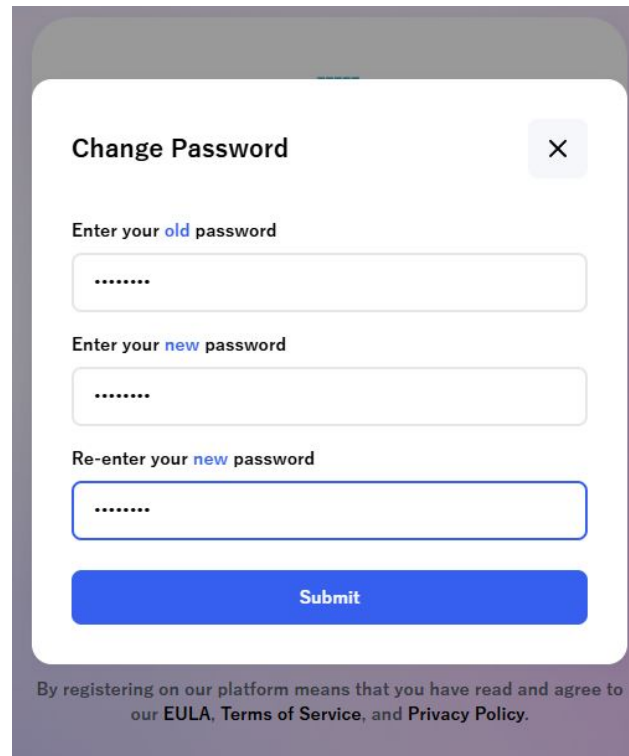
**Glows.AI account for each student.**

**Account:** {student id}@team-7ne8lq6z.glows.ai

**Password:** {student id}

**only 1 credit**

**Please release instance when you don't want to use the Glows.AI**



A screenshot of a 'Change Password' form. The form has a title 'Change Password' and a close button (X) in the top right corner. It contains three input fields: 'Enter your old password', 'Enter your new password', and 'Re-enter your new password'. Each field has a password mask (dots). Below the fields is a blue 'Submit' button. At the bottom of the form, there is a disclaimer: 'By registering on our platform means that you have read and agree to our EULA, Terms of Service, and Privacy Policy.'

# Create instance

The screenshot shows the 'Create instance' page on the Glows.ai platform. The interface includes a sidebar with navigation links, a top header with user information, and a main content area for selecting instance configurations. Red annotations with numbers 1 through 6 highlight specific UI elements: 1 points to the 'Create New' button, 2 points to the 'All' filter dropdown, 3 points to the 'Choose either one' text above the instance table, 4 points to the 'Snapshot' tab, 5 points to the instance details card, and 6 points to the 'Complete Checkout' button.

**Glows.ai** Team 257.060 Credits admin

Home Create New 1 Instances Auto Deploy Datadrive Snapshots Storage Space Billing Profile Member Permissions & Quota Team Setting

GPU CPU ☐ vGPUs All 2 Global

| GPU Model                                                | GPUs | GPU RAM | Region | Avail Unit | vCPUs | RAM   | Price/h      |
|----------------------------------------------------------|------|---------|--------|------------|-------|-------|--------------|
| <input type="radio"/> NVIDIA GeForce RTX 4090            | 1    | 24 GB   | TW-03  | 10         | 10 ⓘ  | 50 GB | 0.490 Credit |
| <input type="radio"/> NVIDIA GeForce RTX 4090            | 1    | 24 GB   | TW-03  | 6          | 10 ⓘ  | 64 GB | 0.550 Credit |
| <input checked="" type="radio"/> NVIDIA GeForce RTX 4090 | 1    | 24 GB   | TW-03  | 8          | 10 ⓘ  | 64 GB | 0.560 Credit |

3. Choose either one

Official My Images 4 Snapshot Selected -

5 snp-e6d7186w Team

VNC-isaacsim-physical-umi

Created at 2026-02-10 12:22

By admin@team-7ne8lq6z.glows.ai

Price/h 0.560 Credit Instance Ordered 1 Show Detail 6 Complete Checkout

# Port Forwarding

- Once the instance has been created, ensure that port forwarding is configured for ports **5901** and **8888** for subsequent use.

The screenshot displays the Glows.ai dashboard interface. On the left, a sidebar contains navigation links: Home, Create New, Instances (highlighted with a red '1'), Auto Deploy, Datadrive, Snapshots, Storage Space, Billing, Profile, Member, Permissions & Quota, and Team Setting. The main content area is titled 'My Instances' and includes tabs for 'Member View' and 'Admin View'. A 'Create New' button is in the top right. Below the tabs, there are filters for 'Instance Status' (Running, Snapshotting, Released) and 'Region' (Global). A table lists instances, with the first instance 'ins-zgvo17or' (NVIDIA GeForce RTX 4090) selected. Below the table, there are tabs for 'Access', 'Monitor', 'Billing', 'Config', 'Datadrive', and 'Hardware'. The 'Access' tab is active, showing 'SSH Port 22', 'TCP Port 5901', and 'TCP Port 8888'. An 'SSH Command' is provided: 'ssh -p 25617 glows@tw-05.access.glow.ai'. A 'New Port Forwarding' button (labeled with a red '4') is at the bottom left. A modal window titled 'New Port Forwarding Rules' is open on the right. It has a 'Service Port\*' input field (labeled with a red '5') and a dropdown menu (labeled with a red '3'). Below the input field, a message states: 'Your port must be a number between 1 to 65535.' The 'Protocol' section has 'TCP' selected and 'HTTPS' as an option. At the bottom right of the modal are 'Cancel' and 'Create' buttons (labeled with a red '6').

# Usage

- **Ensure that an instance has been spawned on GlowsAI.**
- Input: `{hostname}:{Port}` / Password: “password”

My Instances Member View Admin View Create New

Instance Status: Running Snapshotting Released Region: Global

| ID           | Name                    | Region | Status  | Cost          | Action                              |
|--------------|-------------------------|--------|---------|---------------|-------------------------------------|
| ins-zgvoj7or | NVIDIA GeForce RTX 4090 | TW-03  | Running | 1.760 Credits | <span>Release</span> <span>⋮</span> |

Access Monitor Billing Config Datadrive Hardware

SSH Port 22 TCP Port 5901 TCP Port 8888

Hostname: tw-05.access.glows.ai 🔗 Port: 25353 🔗

🔗 New Port Forwarding 🗑 Delete This Port

New TightVNC Connection

Connection

Remote Host: tw-05.access.glows.ai:25353 Connect Options...

Enter a name or an IP address. To specify a port number, append it after two colons (for example, mypc:15002).

Reverse Connections

Listening mode allows people to attach your viewer to their desktops. Viewer will wait for incoming connections. Listening mode

TightVNC Viewer

TightVNC is cross-platform remote control software. Its source code is available to everyone, either freely (GNU GPL license) or commercially (with no GPL restrictions).

Version info... Licensing Configure...

# Installation Notes for VNC Clinet

**We prepare for three operating system user different guideline**

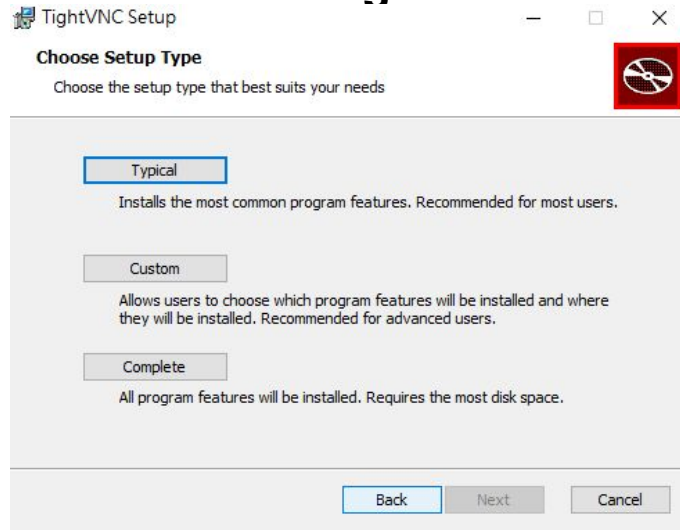
- **Windows**
- **Linux**
- **MacOS**

# Installation Notes (Windows Only)

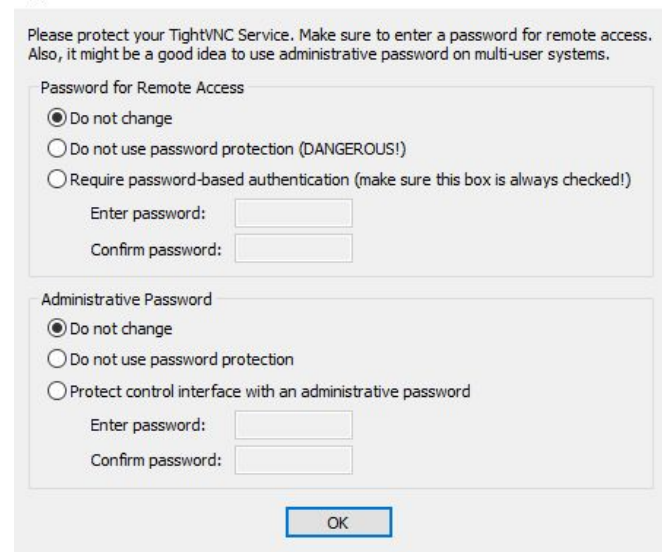
- The following installation slides are intended for **Windows** users.
  - Install TightVNC [[website](#)]
- Users on **Linux** or **macOS** should skip this section.

# Installation Notes (Windows Only)

- During the installation process, please select “**Typical**” installation.
- Additionally, please ensure that both the upper and lower options are set to “**Do not change.**”

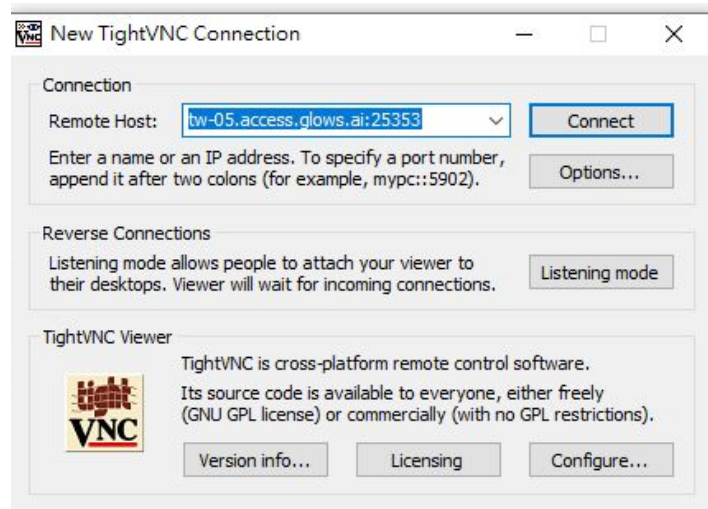
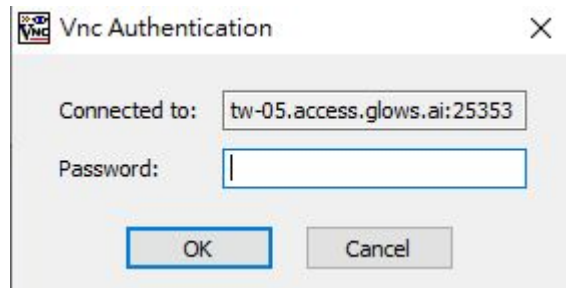
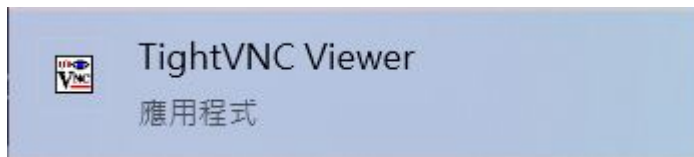


TightVNC Server: Set Passwords



# Installation Notes (Windows Only)

- Finally, open **TightVNC Viewer**, enter the **Remote Host**, click **Connect**, and enter the password.



# Installation Notes (Linux Only)

Users on **Windows** or **macOS** should skip this section.

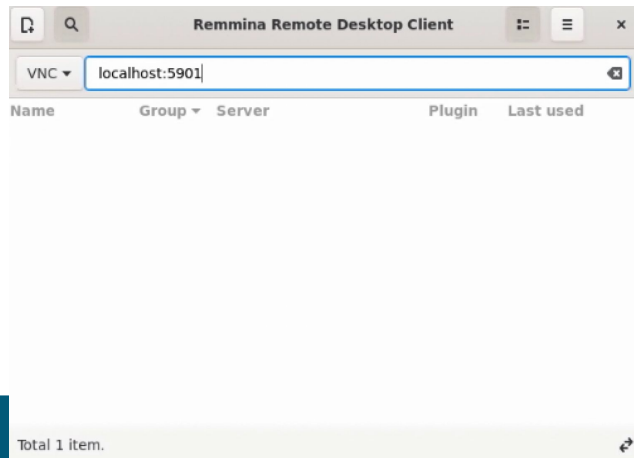
**Please follow the steps:**

**Install remmina command**

1. `sudo apt update`
2. `sudo apt install remmina remmina-plugin-rdp remmina-plugin-secret`

**Login**

3. Input: `localhost:5901`
4. Password: “password”

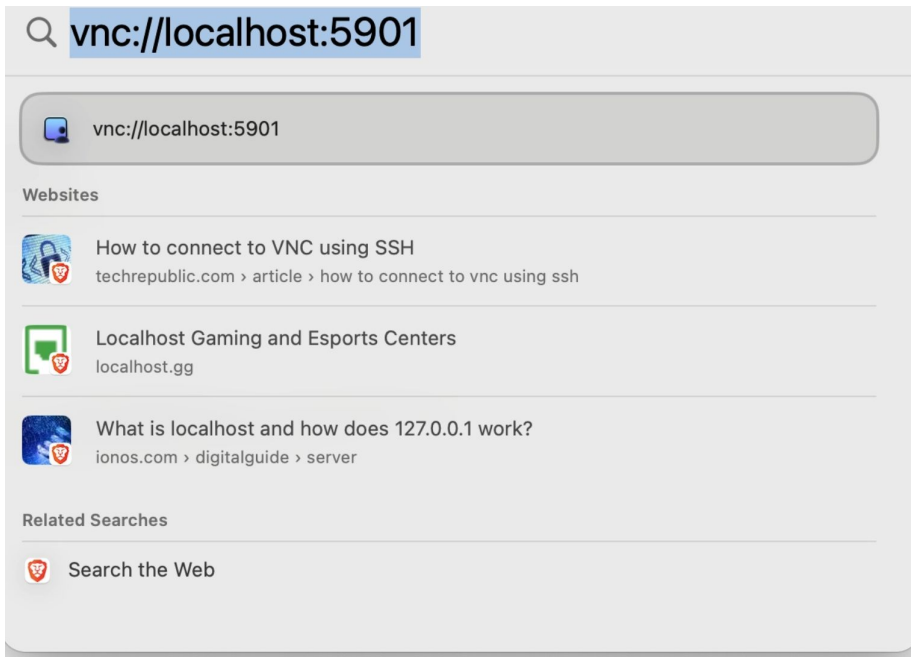


# Installation Notes (MacOS Only)

Users on **Windows** or **Linux** should skip this section.

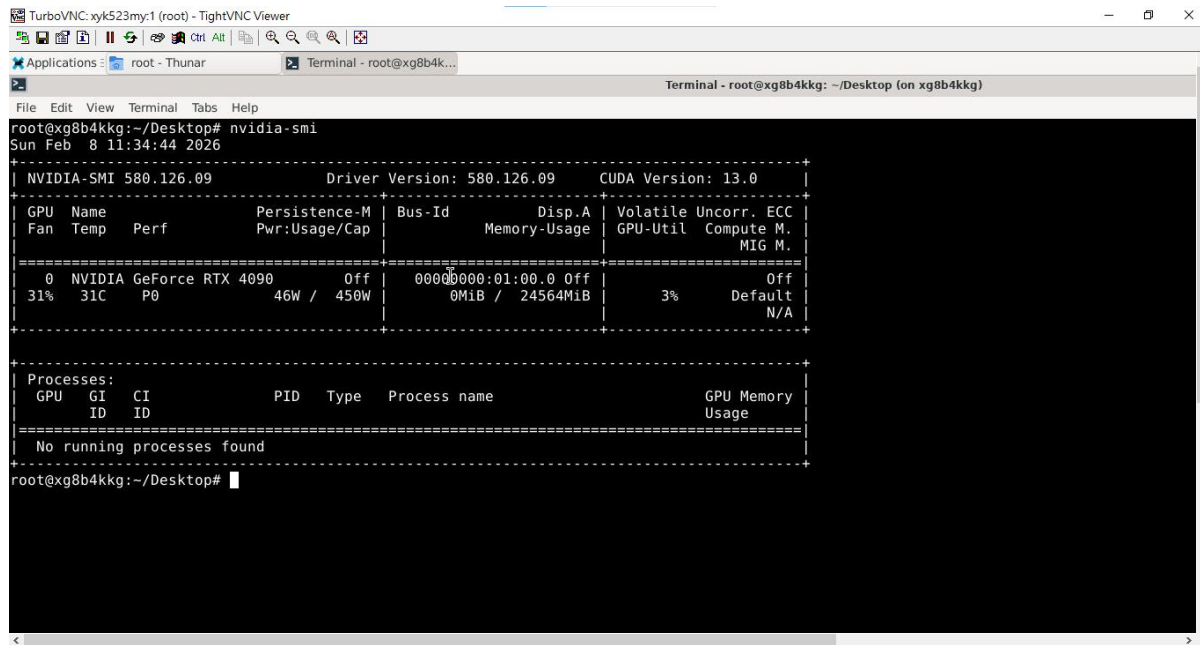
**Please follow the steps:**

1. Shortcut: Command + Space
2. Input: ``vnc://localhost:5901``
3. Password: “password”



# Testing on your VNC Clinet

Please test your machine have GPU via VNC client connection interface



The screenshot shows a VNC client window titled "TurboVNC: xyk523my:1 (root) - TightVNC Viewer". The window contains a terminal application with the following content:

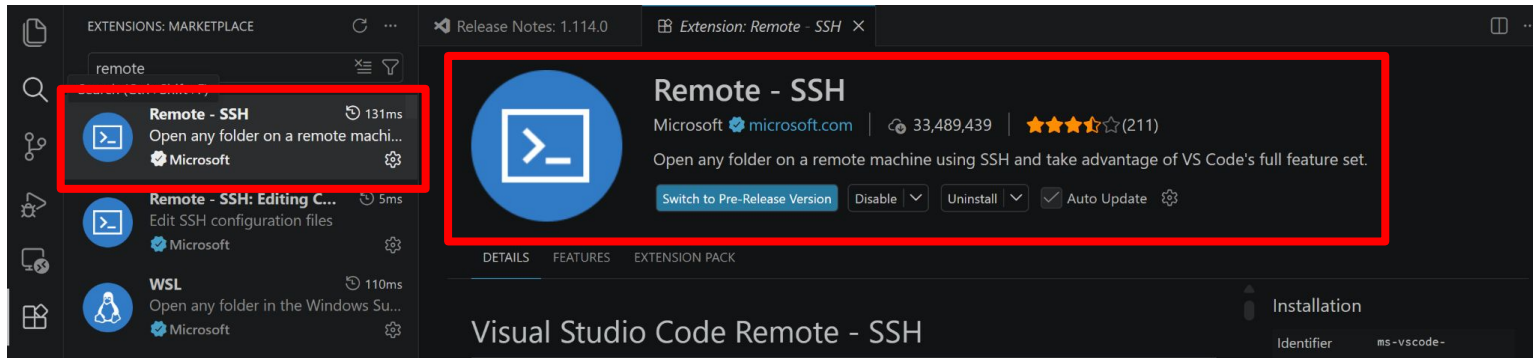
```
root@xg8b4kkg:~/Desktop# nvidia-smi
Sun Feb  8 11:34:44 2026
```

| NVIDIA-SMI 580.126.09 |                         | Driver Version: 580.126.09 |                  | CUDA Version: 13.0 |                      |
|-----------------------|-------------------------|----------------------------|------------------|--------------------|----------------------|
| GPU                   | Name                    | Persistence-M              | Bus-Id           | Disp.A             | Volatile Uncorr. ECC |
| Fan                   | Temp                    | Perf                       | Memory-Usage     | GPU-Util           | Compute M.           |
|                       |                         | Pwr:Usage/Cap              |                  |                    | MIG M.               |
| 0                     | NVIDIA GeForce RTX 4090 | Off                        | 00000000:01:00.0 | Off                | Off                  |
| 31%                   | 31C                     | P0                         | 46W / 450W       | 0MiB / 24564MiB    | 3% Default           |
|                       |                         |                            |                  |                    | N/A                  |

```
Processes:
GPU  GI  CI          PID  Type   Process name                      GPU Memory
  ID  ID                               Usage
=====
No running processes found
root@xg8b4kkg:~/Desktop#
```

# Testing VScode Connection with glows.ai server

Please test your local machine to be able to connect to the glows.ai machine via Remote - ssh extension in the VScode



# Testing VScode Connection with glows.ai server

1. Please open connection with ctrl+shift+P
2. Remote-SSH: Connect to Host...
3. Configure SSH Hosts...
4. C:\Users\user\.ssh\config

Enter your glows ai information into the Connection setting file example:

```
C: > Users > user > .ssh > ≡ config  
1 Host tw-07.access.glowes.ai  
2   |   HostName tw-07.access.glowes.ai  
3   |   Port 25967  
4   |   User root
```

# Testing VScode Connection with glows.ai server

## Download the codebase for homework4

1. `git clone https://github.com/HCIS-Lab/ai-capstone26.git`
2. `cd ai-capstone26/hw4`

You can start to programming in the VScode with **fk.py** and **ik.py** .

# Summary

1. **Connection with VScode**
  - a. for coding your homework4 fk.py and ik.py program.
2. **Connection with VNCserver**
  - a. for testing your fk.py and ik.py program and visualizing your result.

Congratulation! You can start to coding for your Homework4~